

Review

Applying utility values in cost-effectiveness analyses of lung cancer screening: A review of methods

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ABSTRACT

Lung cancer screening with low-dose computed tomography (LDCT) in high-risk populations has been shown in randomised controlled trials to lead to early diagnosis and reduced lung cancer mortality. However, investment into screening will largely depend on the outcomes of cost-effectiveness analyses that demonstrate acceptable costs for every quality-adjusted life year (QALY) gained. The methods used to apply utility values to measure QALYs can significantly impact the outcomes of cost-effectiveness analyses and if applied inaccurately can lead to unreliable estimates.

We reviewed the use of utility values in 26 cost-effectiveness analyses of lung screening with LDCT conducted between 2005 and 2021, and found considerable variation in methods. Specifically, authors made different assumptions made relating to (i) baseline quality-of-life among screening participants, (ii) potential harms from screening, (iii) utilities and disutilities applied to lung cancer health states, and (iv) quality-of-life for lung cancer survivors. We discuss how each of these assumptions can influence incremental cost-effectiveness ratios.

Key recommendations for future evaluations are (i) that modelling studies should justify the choice of baseline utilities, especially if patients are assumed to recover fully after curative treatment; (ii) the impact of false positive scans on quality-of-life should be modelled, at least in sensitivity analyses; (iii) modellers should justify assumptions relating to post-operative recovery, preferably based on knowledge of local practices; (iv) utilities applied to a lung cancer diagnosis should be appropriately sourced and calculated; and (v) adjustment for age-related declines in quality-of-life should be considered, especially for models that examine lifetime horizons.

1. Introduction

Lung cancer screening with low dose computed tomography (LDCT) is a priority issue in lung cancer control for many countries. Randomised controlled trials have demonstrated that targeted screening can reduce lung cancer-specific mortality in high-risk groups, namely older adults with a history of heavy smoking [1–3]. However, national programmatic lung cancer screening has yet to be widely adopted. Health economic evaluations of the balance of costs and health benefits are key to health-system policy and investment decisions. On current evidence, cost-effectiveness estimates of lung cancer screening vary greatly across settings [4]. Variability in cost-effectiveness estimates can arise from

differences in country-specific input parameters, but also due to the underlying assumptions of modelled analyses and choice of data inputs.

A key metric in cost-effectiveness analyses is the quality-adjusted life year (QALY). QALYs capture the impact of an intervention on both the duration of life and quality-of-life. Dividing the cost of an intervention by the expected gains in QALYs gives an incremental cost-effectiveness ratio (ICER), which serves as a summary measure of cost-effectiveness. QALYs are calculated as the product of the duration of a health state and a numerical weight, known as a utility. Utility values typically range from 0.00 (dead) to 1.00 (perfect health), though negative utilities (indicating states worse than death) are also possible [5]. Utilities are elicited directly with techniques such as the standard gamble and time

Abbreviations: QALY, quality-adjusted life year; LDCT, Low-dose computed tomography; MAUI, multiple-attribute utility instrument.

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Table 1
Summary of cost-utility analyses of lung cancer screening.

First author	Year	Setting	Analytical approach	Time horizon	Instrument	ICER versus no screening (cost per QALY)
Manser [10]	2005	Australia	Markov model	15 years	Investigator-rated [†]	AUD 105,090
McMahon [11]	2011	United States	Microsimulation	Lifetime	EQ-5D, investigator-rated [†] , rating scale [‡]	Men: USD 149,000 Women: USD 137,000
Shmueli [12]	2013	Israel	Decision tree	Lifetime	EQ-5D	USD 1,464
Villanti [13]	2013	United States	Actuarial analysis	15 years	SF-6D, meta-analysis [36]	USD 28,240
Black [14]	2014	United States	QALE comparison	Lifetime	SF-6D	USD 81,000
Goffin [15]	2015	Canada	Microsimulation	20 years	HUI-3 and CLAMES	CAD 52,000
Evans [16]	2016	Canada	Microsimulation	20 years	HUI-3 and CLAMES	No ICER calculated
Field [17]	2016	United Kingdom	LE comparison	Lifetime	EQ-5D	GBP 8466
Goffin [18]	2016	Canada	Microsimulation	20 years	HUI-3 and CLAMES	Annual: CAD 52,000; Biennial: CAD 30,000–51,000
Cressman [19]	2017	Canada	Markov model	30 years	EQ-5D	CAD 20,724
Yang [20]	2017	Taiwan	QALE comparison	Lifetime	EQ-5D	USD 19,683
Hofer [21]	2018	Germany	Microsimulation	15 years	EQ-5D, investigator-rated [†] , meta-analysis [36]	EUR 30,291
Jaine [22]	2018	New Zealand	Macrosimulation	Lifetime	Disability weights	USD 104,000
Kumar [23]	2018	United States	Multistate model	Lifetime	Meta-analysis [36]	USD 60,000
Snowsill [24]	2018	United Kingdom	Microsimulation	Lifetime	EQ-5D	GBP 28,000 [§]
Wade [25]	2018	Australia	QALE comparison	10 years	SF-6D	AUD 233,000
Criss [26]	2019	United States	Microsimulation	45 years	SF-6D, EQ-5D, and unclear [‡]	NLST: USD 49,200; CMS: USD 68,600; USPSTF: USD 96,700
Toumazis [32]	2019	United States	Microsimulation	Lifetime	EQ-5D [‡] , unclear ^{‡, #}	USPSTF: USD 96,213; CMS: USD 90,702
Evans [27]	2020	Canada	Microsimulation	Lifetime	HUI-3 and CLAMES	CAD 22,200
Veronesi [28]	2020	Italy	Decision tree	5 years	Investigator-rated [†]	EUR 3,297
Cadham [29]	2021	United States	Microsimulation	Lifetime	SF-6D, EQ-5D, unclear [‡]	No comparisons to “no screening” scenario
Esmaili [35]	2021	Iran	Markov model	Unclear	Unclear [‡]	IRR 98,515,014
Gomez-Carballo [30]	2021	Spain	Markov model	Lifetime	Meta-analysis [36], vignette-based SG [40], unclear [‡]	EUR 2345
Kim [33]	2021	South Korea	Markov model	45	Unclear ^{††} , vignette-based SG [41]	USD 25,383
Sun [31]	2021	China	Markov model	Lifetime	Meta-analysis [36]	USD 13,473–15,736
Yuan [34]	2021	China	Markov model	50	EQ-5D, meta-analysed [37]	No ICER reported

Abbreviations: AUD, Australian dollars; CAD, Canadian dollars; CLAMES, Classification and Measurement System of Functional Health; CMS, Centre for Medicare and Medicaid Services; EUR, Euro; EQ-5D, EuroQol-5-Dimension; SF-6D, Short-form Six-Dimension; GBP, pound sterling; HUI-3, Health Utilities Index 3; ICER, incremental cost effectiveness ratio; IRR, Iranian rial; NLST, National Lung Screening Trial; QALE, Quality-adjusted life expectancy; QALY, quality-adjusted life year; SF-6D, Short-form Six-Dimension; SG, standard gamble; USD, United States dollars; USPSTF, United States Preventive Services Taskforce.

[†] Utilities transformed into standard gamble equivalents by Earle et al. [42].

[‡] Exact utilities could not be identified in cited reference.

[§] Single screen for adults aged 60–75 with 3% risk of lung cancer versus no screening.

[#] Values identified in referenced paper but did not match original source.

^{††} Cited source could not be accessed.

trade-off, or indirectly using multiple attribute utility instruments (MAUIs). Examples of MAUIs include the EuroQol 5 Dimension (EQ-5D) questionnaire [6] and the Short-form Six-Dimension (SF-6D) [7]. Checklists for searching, reviewing, and synthesising utilities are available and comprehensive [8,9] but guidance on their application is scarce. Recommendations on using utilities in health economic models have been provided by a recent ISPOR Good Practice Report [9] but, due to the general scope of the report, require some elaboration for specific contexts. There are unique considerations to be made for evaluations of cancer screening that merit discussion.

While much attention has been devoted to ensuring the accuracy of screening-related parameters (e.g., efficacy, sensitivity, specificity) in cost-effectiveness evaluations of lung cancer screening, the application of utility values in these models has received less scrutiny. Choice of utility values and assumptions relating to quality-of-life can significantly impact modelled estimates and be a decisive factor in policy recommendations, particularly when ICERs fall near local willingness-to-pay thresholds.

All health economic models are necessarily limited by the availability and quality of the data used for parameterisation and are therefore dependant on key underlying assumptions. We aimed to compare the assumptions around the application and choice of utility weights used in existing cost-effectiveness analyses of lung cancer screening with LDCT, and discuss potential implications of these assumptions for

estimates of QALYs and ICERs. When evaluating lung screening, there are four main considerations in relation to utilities, and we address each in turn: (i) baseline quality-of-life among screening participants, (ii) potential harms from screening, (iii) lung cancer utilities and disutilities, and (iv) quality-of-life for lung cancer survivors.

2. Studies selected

This review covers cost-effectiveness analyses of LDCT screening for lung cancer published between January 2005 and August 2021 identified in Pubmed using the search terms ‘lung’, ‘cancer’, ‘screening’ and ‘cost’, as well as the reference lists of appropriate papers. As the focus of this review was on utility values, studies that did not report QALYs were excluded. If there was more than one publication describing the same analysis, the more detailed version was retained.

Twenty-six cost-effectiveness analyses of LDCT for lung cancer screening that incorporated quality-of-life were identified [10–35]. The details of each analysis are presented in Table 1. These varied in setting, analytic approach, and choice of utility values.

The majority of studies used utilities obtained with MAUIs. Five studies used exclusively EQ-5D utilities [12,17,19,20,24] and two studies used exclusively SF-6D utilities [14,25]. The remaining studies sourced utilities derived with a mixture of instruments and methods. These include Health Utilities Index-3 (HUI-3) values, the Classification

Table 2

Assumptions relating to quality-of-life in cost-utility analyses of lung cancer screening.

First author	Year	Baseline utilities	Tested sensitivity to utilities	Screening harms	Lung cancer utilities	Recovery	Adjustment for ageing
Manser [10]	2005	Perfect health	Yes	Yes	Mean utility	Full	No
McMahon [11]	2011	Age-related norms	Yes	No	Multiplicative (unscaled)	Full	Yes
Shmueli [12]	2013	Perfect health	Partially [†]	No	Mean utility	Full	No
Villanti [13]	2013	Age-related norms	No	No	Multiplicative (unscaled)	No	Yes
Black [14]	2014	Trial (NLST)	Yes	Yes	Mean utility	Partial	Yes
Goffin [15]	2015	Age-related norms	No	No	Multiplicative (unscaled)	Partial	Yes
Evans [16]	2016	Age-related norms	No	No	Multiplicative (unscaled)	Partial	Yes
Field [17]	2016	None	No	No	None [‡]	Full [‡]	No
Goffin [18]	2016	Age-related norms	No	No	Multiplicative (unscaled)	Partial	Yes
Cressman [19]	2017	Trial (PanCan)	Yes	Yes [§]	Mean utility	No	No
Yang [20]	2017	Age- and sex-matched	No	No	Continuous function	Potentially	No
Hofer [21]	2018	Age-related norms	No	No	Disutility (unclear)	No	No
Jaine [22]	2018	Perfect health	No	Yes	Disability weights	Partial	No
Kumar [23]	2018	Perfect health	Yes	No	Mean utility	No	No
Snowsill [24]	2018	History of smoking	Yes	Yes	Additive	Full [¶]	No
Wade [25]	2018	Trial (NLST)	Yes	Yes	Mean utility	Partial	Yes
Criss [26]	2019	Age-related norms	Yes	No	Disutility (unclear)	No	Yes
Toumazis [32]	2019	Age-related norms	Yes	Yes	Unclear [#]	Full	Unclear [#]
Evans [27]	2020	Age-related norms	No	No	Multiplicative (unscaled)	Partial	Yes
Veronesi [28]	2020	Unclear	No	No	Unclear	No	Unclear
Cadham [29]	2021	Age-related norms	No	No	Disutility (unclear)	No	Yes
Esmaeili [35]	2021	Unclear	Yes	No	Mean utility	Full	No
Gomez-Carballo [30]	2021	Unclear	Yes	No	Mean utility	Partial	No
Kim [33]	2021	History of smoking	Yes	No	Unclear [#]	No	Unclear [#]
Sun [31]	2021	Perfect health	No	No	Mean utility	No	No
Yuan [34]	2021	Unclear	Yes	No	Mean utility	No	No

Abbreviations: NLST, National Lung Screening Trial; PanCan, Pan-Canadian Early Detection of Lung Cancer.

[†] Varied quality-adjusted life expectancy in sensitivity analysis but not utilities specifically.[‡] Authors applied mean utility of UK population aged ≥ 65 to incremental QALYs, assuming that incremental QALYs accrued to early-stage lung cancer patients who had the same quality-of-life as the general population. Equivalent to assuming that patients fully recover (with 0 time to recovery).[§] Applied utilities from PanCan trial to screened cohort.[#] Baseline utilities were age-dependent, but unclear if ageing impacted quality-of-life after lung cancer diagnosis.[¶] Authors did not apply disutility to stage I lung cancers. Equivalent to assuming that patients fully recovery after curative treatment (with 0 time to recovery).

and Measurement System of Functional Health (CLAMES), meta-analyses [36,37], disability weights [38,39], vignette-based standard gamble utilities [40,41], and values from a systematic review by Earle et al. [42] who transformed values derived from a rating scale [43] and investigator-rated values [44] to standard gamble equivalents.

3. Utilities applied to individuals eligible for lung cancer screening

Eligibility criteria for lung screening with LDCT focus on older individuals with a history of heavy smoking. In 2013, the US Preventive Services Task Force (USPSTF) recommended LDCT for current and former smokers (who quit in the preceding 15 years) aged 55–80 years with a minimum of 30 pack-years of smoking [45]. These criteria were recently expanded to include adults aged 50–80, with a minimum of 20 pack-years of smoking [46]. Other organisations have endorsed eligibility based on lung cancer risk calculators, in which smoking history is also used [47].

Regardless of the exact criteria used, restricting lung screening to high-risk populations has two implications for cost-effectiveness evaluations. Given that tobacco use is associated with many diseases other than lung cancer, it is expected that populations with a higher risk of lung cancer also have a higher risk of death from other causes [48]. This has been considered in cost-effectiveness analyses. The second implication, which is seldom acknowledged, is that the comorbidities that increase the risk of other-cause mortality also lead to lower quality-of-life. This means that lung screening participants are expected to have lower utility values than the general population. Many economic models assume that screening participants enter a program with either perfect health (i.e., they assume a baseline utility of 1.00), or that they can be represented by age- and sex-specific population norms. Both assumptions result in an overestimate of pre-screening quality-of-life. Ideally, utility values applied at baseline (pre-screening) should reflect the

health profile of a lung screening population.

To date, only 5 analyses used baseline utility values that represent a lung screening population; 3 used lung screening trial data [14,19,25] and 2 used utilities specific to people who smoked [24,33] (Table 2). Of the remaining studies, five assumed perfect health for baseline utilities [10,12,22,23,31] while 10 studies used population norms [11,13,15,16,18,21,26,27,29,32]. One study [20] used utilities from an age- and sex-matched population to estimate loss of quality-adjusted life expectancy (QALE), but did not match by smoking history or comorbidities. One did not apply baseline utilities [17].

4. Potential harms of screening

It is necessary that screening programs deliver more benefits than harms in the target population [49]. Harms can arise through i) radiation exposure, ii) incidental findings, iii) overdiagnosis (where cancers that would not manifest clinically in the lifetime of the screenee are diagnosed) resulting in unnecessary treatment, iv) disutility from invasive tests and procedures for tumours that are ultimately benign, and v) anxiety from false positives and indeterminate findings.

Some studies show temporary distress in lung screening participants who receive a false positive scan. One trial evaluating lung screening found a statistically significant decrease in EQ-5D utilities in participants who were notified of an actionable nodule [50]. In the UK Lung Screening (UKLS) trial, cancer distress (measured with the Cancer Worry Scale) and anxiety (measured with the Hospital Anxiety and Depression Scale) were also elevated after notification of a positive result but this did not persist in the long-term [51]. Furthermore, indeterminate findings in the NELSON trial were associated with higher anxiety in the State Trait Anxiety Inventory (STAI) and lower EQ Visual Analogue Scale (VAS) scores compared to negative screening results, though no effect was identified in SF-12 physical and mental component scores (PCS and MCS) [52].

Other studies have not identified an association between screening results and quality-of-life. Taghizadeh et al. [53] measured the EQ-5D, EQ VAS, STAI, and the SF-12 PCS and MCS among participants of the PanCan study and found no changes in quality-of-life in those notified of a positive CT scan. Likewise, participants in the NLST who received negative and false positive scans reported similar quality-of-life outcomes, as measured with the STAI and SF-36 PCS and MCS [54].

Any cost-effectiveness evaluation of an intervention should account for its potential harms. However, in the case of lung cancer screening, it is unclear how harms related to quality-of-life should be quantified. The weight of the evidence suggests no substantial change in utility after a positive screen, with only one study [50] showing a statistically significant decline (and only among 25 individuals). However, it is possible that generic instruments are not sensitive to the psychological effects of screening. Measures of cancer distress and anxiety may be more meaningful but cannot be easily mapped to preferences. In the absence of a clear solution, a reasonable approach may be to test a range of values through sensitivity analysis. This range may include, on one end, the optimistic assumption of no decline in quality-of-life and, on the other, the conservative assumption that positive scans induce a quality-of-life equivalent to an asymptomatic lung cancer diagnosis until the next scan.

Of the 26 cost-effectiveness analyses identified, 7 had accounted for the possible effects of screening on participants' quality-of-life, either in the base case analysis or in a sensitivity analysis [10,14,19,22,24,25,32] (Table 2). Four of these studies considered the effects of anxiety due to false positives and/or indeterminate findings [14,24,25,32], 2 considered the effects of unnecessary diagnostic procedures on quality-of-life [21,22], and 1 considered the effects of both [10]. One of these studies applied different, trial-derived utilities to individuals who were screened [19]. One study [24] assumed that participants with a false-positive scan would experience a short-term disutility exceeding that assumed for a lung cancer diagnosis.

5. Utilities applied at lung cancer diagnosis

Different methods exist to capture the effects of lung cancer on quality-of-life. The simplest approach is to apply a fixed value representing the mean utility of lung cancer patients, though this can lead to illogical combinations of inputs. For example, Hanmer et al. reported a mean EQ-5D utility of 0.724 for women aged 80–89 [55] which is lower than the mean EQ-5D utility for stage I lung cancer (mean = 0.77) reported by Tramontano et al. [56]. If these values were naively combined in a model without adjustment, elderly lung cancer patients would see their quality-of-life improve after diagnosis.

5.1. Disutilities

One solution is to apply disutilities that capture the quality-of-life burden imposed by lung cancer. By applying a modifier to baseline values across all individuals (e.g., all age groups), modellers can maintain a logical hierarchy of health states in their evaluation. This can also be performed in reverse where a mean lung cancer utility is applied first, then modified for ageing.

Disutilities can be additive or multiplicative. The former takes the form of an absolute *decrement* which is subtracted from a baseline value while the latter is a scaled *multiplier* used to reweight baseline values [57]. To calculate a utility decrement, the disease-specific score is subtracted from a relevant comparison value. For example, a diabetes-related utility of 0.70 should be subtracted from a comparison utility (representing diabetes-free individuals) of 0.80 to yield a decrement of 0.10. A multiplier, on the other hand, should be calculated as the diabetes-specific value divided by the comparison value. With the previous example, a scaled multiplier for diabetes would be calculated as $0.70/0.80 = 0.875$.

The absolute change in utility can differ between the additive and multiplicative method. Whereas decrements will always have the same

effect in absolute terms, multipliers produce larger reductions when the baseline value is high and smaller reductions when the baseline is low. Applying the disutilities calculated above to a baseline value of 0.60 would yield scores of 0.50 ($0.60 - 0.10 = 0.50$) under the additive method and 0.525 ($0.60 \times 0.875 = 0.525$) under the multiplicative method. If a baseline of 0.90 were used, the corresponding utilities would be 0.80 and 0.788. It should be noted that in certain cases, disutilities can produce values outside the range allowed by a scoring system. The SF-12 variant of the SF-6D, for example, has a floor of 0.345 when derived using the algorithm by Brazier et al. [58]. The additive method can produce scores below this value and, indeed, below 0. In contrast, the multiplicative method cannot produce negative values but may still lead to utilities below the allowed range.

When calculating disutilities, the choice of comparison group is crucial. Disutilities should reflect the burden imposed by a factor of interest, relative to the same population *without* the factor of interest. When the effect of an intervention is needed (e.g., the direct impact of screening), disutilities are best acquired from randomised controlled trials where confounders are distributed equally between study arms. For disease-related disutilities, estimates from regressions or comparisons with matched controls are suitable alternatives. These must control for other determinants of quality-of-life, particularly comorbidities. Where such options are not available, comparisons with age-related norms can be used, per the additive and multiplicative methods described above. This last approach is valid but warrants caution. If the at-risk population has a higher prevalence of comorbidities than the general population, the resulting disutilities will overestimate the impact of the disease. As argued earlier, the population at risk of lung cancer bears a heavier burden of tobacco use and should not be substituted by the general population when calculating disutilities.

5.2. Lung cancer utilities in the cost-effectiveness literature

There was considerable variation in the calculation of lung cancer utilities (Table 2). Mean values were applied in 10 studies [10,12,14,19,23,25,30,31,34,35], additive disutilities were applied in one study [24], and multiplicative disutilities were applied in 6 studies [11,13,15,16,18,27]. One study [22] applied disability weights as a substitute for disutilities. One study applied time-dependent utilities in the form of a smoothed function to estimate QALE [20]. Three studies applied lung cancer-related disutilities but did not specify the method used [21,26,29]. Two studies reported values used to represent lung cancer health states, but did not state if they were applied directly or used to modify age-related baseline scores which were also reported [32,33]. Where multiplicative adjustment was performed [11,13,15,16,18,27], disutilities were not scaled. Rather, lung cancer-related utilities were multiplied with age-related norms, producing values that potentially underestimated the quality-of-life of lung cancer patients.

6. Utilities applied to lung cancer survivors

The benefits of screening are determined primarily by the outcomes of patients with operable disease, particularly stage I lung cancer. Thus, assumptions relating to the quality-of-life of lung cancer survivors can be consequential.

6.1. Recovery

Existing evidence suggests that patients experience a substantial decline in physical health after surgery that may last up to two years [59]. Differences in outcomes can arise from variations in medical procedures and *peri-operative* care. Pneumonectomy (the removal of a lung) is associated with poorer post-surgical quality-of-life compared to lobectomy (the removal of a lobe) [60]. Minimally invasive procedures such as video-assisted thoracoscopic surgery (VATS) are associated with

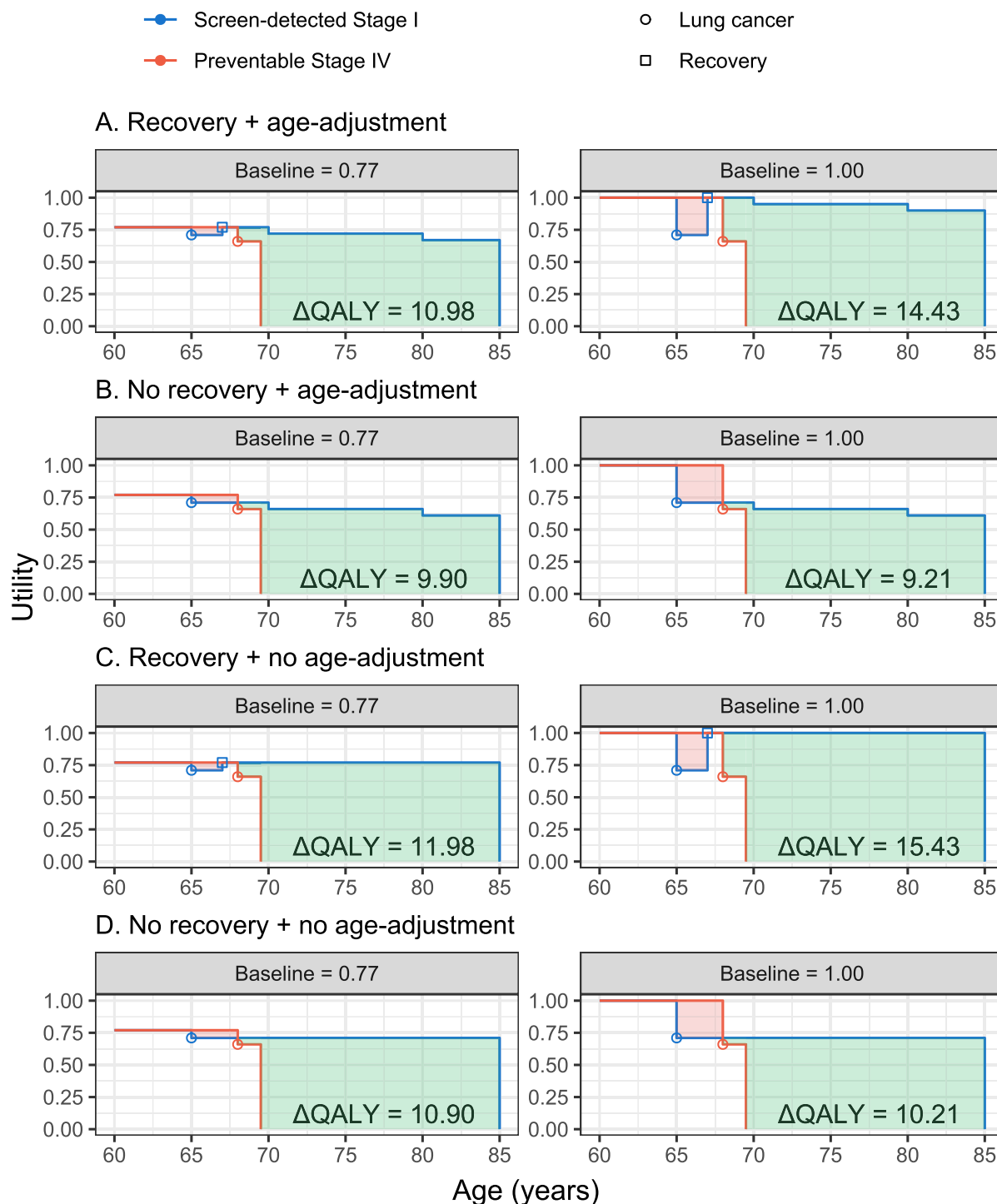


Fig. 1. Utility trajectories of identical individuals diagnosed with lung cancer under varying assumptions relating to baseline utility, recovery, and ageing. The screened individual (blue) is diagnosed with stage I lung cancer at age 65 (utility (U) = 0.71), fully recovers at age 67, deteriorates due to ageing (0.05 decrement at ages 70 and 80), and dies at 85. The unscreened individual (red) is diagnosed with stage IV disease at 68 (U = 0.66) and lives for 18 months. Differences in quality-adjusted life years gained ($\Delta QALY$) by screening is shown under scenarios in which the screened individual (A) experiences full recovery and utilities decline with age, (B) does not return to their original quality-of-life and utilities decline with age, (C) recovers and utilities do not decline with age, (D) does not recover and utilities do not decline with age. Each scenario is represented assuming either perfect health at baseline (right panels) or less than perfect health at baseline (left panels). Utility values and event times were selected to be illustrative, rather than representative.

fewer complications and may improve quality-of-life over thoracotomy [61]. Further, implementation of enhanced recovery after surgery (ERAS) protocols may be important in facilitating patient recovery [62]. Thus, variations in care should be considered when populating cost-effectiveness models. Knowledge of local practices may be necessary to inform model design.

Of the studies identified, 4 assumed that patients returned to their baseline quality-of-life [10–12,32], while 8 assumed that patients returned to an intermediate quality-of-life [14–16,18,22,25,27,30] (Table 2). One study assumed that there was no impairment from surgery on the grounds that the baseline utility chosen for their model was lower than the utilities they had identified for stage I lung cancer [24]. One study assumed that the life-years gained from screening would accrue to cured patients, to whom they applied population norms [17]. One study [19] assumed no recovery after lung cancer treatment. Yang et al. [20] applied utilities as a function of time since diagnosis. These were stratified by stage and therefore could have captured post-operative changes to quality-of-life, though it is unclear if recovery was observed in their data.

The remaining studies surveyed did not explicitly report modelling decisions relating to post-operative recovery, but their assumptions were deduced based on other information available. One study [35] used a Markov model in which individuals with curable lung cancer could transition to a ‘healthy’ state, and was thus assumed to have implemented an improvement in quality-of-life. Five studies [13,26,28,29,33] did not report parameters relating to the duration of post-operative or stage I lung cancer health states and were, therefore, assumed to have not modelled changes in quality-of-life. In addition, Hofer et al. applied the same utilities for stage I lung cancer aftercare as they did for surgery, indicating no change in quality-of-life over time [21]. Finally, based on model structure, we deduced that 3 studies did not assume post-operative recovery [23,31,34].

6.2. Age-related quality-of-life

Following the post-operative period, patients who are cured can live for an extended number of years. Over this time, it is expected that their health and quality-of-life will deteriorate due to ageing.

Of the 26 studies identified, 8 applied lung cancer disutilities to population norms [11,13,15,16,18,26,27,29] while two applied an age-related disutility to mean lung cancer scores [14,25] (Table 2). Two studies [32,33] listed age-related baseline utilities but did not state if individuals’ quality-of-life with (or after) lung cancer was affected by ageing. Thirteen studies did not account for age-related decline [10,12,17,19–24,30,31,34,35]. Yang et al. [20] applied utilities as a continuous function of time based on longitudinal EQ-5D data collected over a three-year period. Utilities beyond observed values were extrapolated from the last 10% of observations. Due to the short observation period, age-related changes to quality-of-life could not have been captured. One study provided insufficient information to determine assumptions relating to age-related quality-of-life [28].

Capturing the effects of ageing is only necessary in studies with longer time horizons. Of the 26 studies examined in this review, 13 applied a lifetime time horizon. Of these, 4 adjusted utility values for age [11,14,27,29], 8 did not account for ageing [12,17,20,22–24,30,31], and 1 study [32] was unclear with respect to its assumptions about age-related quality-of-life with (or after) lung cancer.

7. Implications of utility-related modelling assumptions for incremental QALYs

We found substantial variation in the methods used to calculate QALYs in the cost-effectiveness literature. However, it is important to note that only *incremental* benefits will affect cost-effectiveness outcomes. Below, we discuss the impact of each modelling assumption described above on the calculation of incremental QALYs and, by

extension, ICERs.

7.1. Choice of baseline utility

Assuming perfect health or applying population norms in quality-of-life at baseline (pre-screening) will bias the calculation of QALYs but depending on other assumptions in a model, may have little impact on cost-effectiveness. This is because QALYs accrued *prior* to cancer will cancel out between screening and comparator scenarios (ignoring the effects of false positives and indeterminate findings). This also implies, however, that baseline values can be consequential if applied to events after diagnosis.

Early detection incurs a penalty to quality-of-life that can be exaggerated if baseline utilities are overestimated. An illustration of this is provided in Fig. 1, which depicts the hypothetical quality-of-life trajectories of two identical individuals diagnosed with lung cancer, where one is screen-detected and one is not, and compares incremental QALYs in relation to different assumptions around baseline utilities, age-related utilities, and survival. With early diagnosis, the screened individual receives lung surgery at a time when they would have otherwise been asymptomatic. This results in a short-term QALY deficit (indicated by the red-shaded region) that is recouped through long-term survival (indicated by the green-shaded region). If a baseline utility of 1.00 (perfect health) is assumed, the decline in quality-of-life that follows a screen-detected diagnosis is larger than if a value less than 1 is assumed (Fig. 1). As a result, the QALYs gained by screening are underestimated.

Assumptions relating to post-operative recovery for lung cancers detected early will have a larger influence on ICERs. Models that assume a full recovery to baseline will predict greater benefits from screening. This assumption can be justified but, where it is made, it is essential that the baseline utility values selected are appropriate. Fig. 1A and 1C demonstrate this problem. In the examples given, a return to perfect health (utility = 1.00) will overestimate the benefits of screening compared to the more conservative assumption that patients return to a baseline of 0.77.

The time required to recover from surgery should also be considered. Shorter recovery periods lead to higher incremental QALYs. This was explicitly explored in the study conducted by McMahon et al. [11]. In their base case analysis, it was assumed that patients experienced a reduction in quality-of-life in the 3-months following surgery, after which they returned to baseline. This yielded an ICER of USD 149,000/QALY. In a sensitivity analysis, they extended this period to 24 months, causing the ICER to increase to USD 161,000/QALY. When they modelled no recovery, the ICER increased further to USD 209,000/QALY.

7.2. Harms from screening

The benefits of screening accrue to the individuals who are diagnosed with early-stage disease that would have otherwise been detected at an advanced stage. Meanwhile, potential harms accrue to those who do not develop lung cancer but receive a false positive scan or indeterminate finding. False positive rates vary across studies and programs but often outweigh the number of true positives. Hence, the aggregated disutility from false positive scans can have a large impact on cost-effectiveness if they are assumed to be frequent.

The influence of false positives on ICER estimates was demonstrated in several of the studies identified. In a sensitivity analysis, Manser et al. [10] found that screening was less cost-effective when a decrement of 0.04 was applied to every positive scan than when no decrement was applied at all (AUD 131,960/QALY versus AUD 87,311/QALY respectively). In addition, Black et al. [14] estimated an ICER of USD 81,000/QALY in their base case analysis, where no impact of positive screens was assumed, but found that a decrement of 0.05 led to a substantially higher ICER of USD 116,000/QALY. Similarly, Wade et al. [25] found that a decrement of 0.05 led to a doubling of the ICER relative to the base

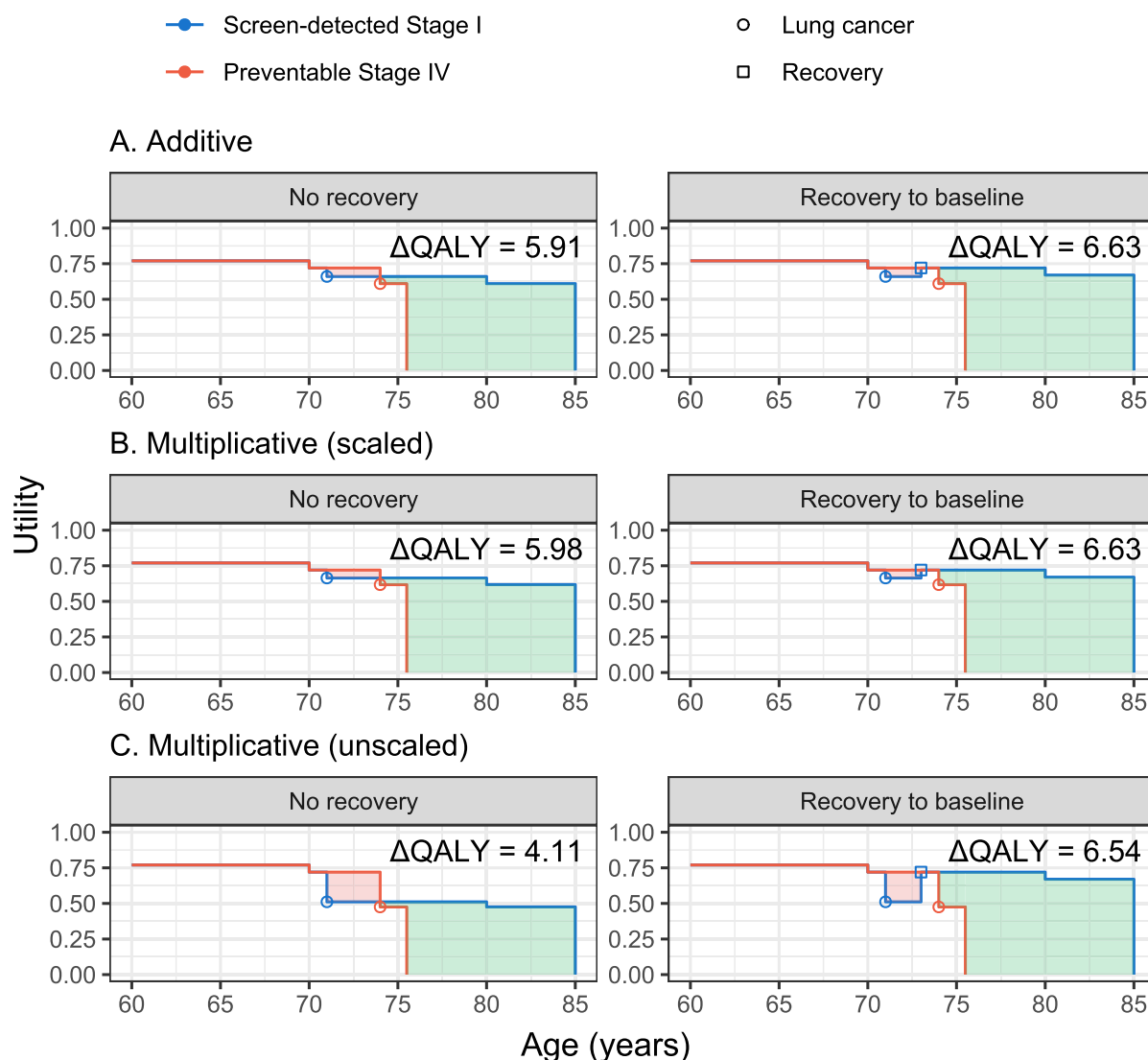


Fig. 2. Utility trajectories of identical individuals diagnosed with lung cancer under different methods for modelling lung cancer-related utility. The screened individual (blue) is diagnosed with stage I disease at 71, recovers at 73, and dies at 85. The unscreened individual (red) is diagnosed with stage IV disease at 74 and lives for 18 months. Disutilities for lung cancer are applied to age-related baseline scores (0.77 at 60, 0.72 at 70, and 0.67 at 80). Differences in incremental quality-adjusted life years (QALYs) are shown when (A) a constant decrement for stage I (0.06) and stage IV (0.11) is applied, (B) a scaled multiplier for stage I (0.922) and stage IV (0.857) disease is applied, (C) baseline scores are multiplied directly by utilities for stage I (0.71) and stage IV (0.66). Each scenario is represented assuming either a return to the pre-screening health state at baseline (right panels) or no recovery in quality-of-life after diagnosis (left panels). Utility values and event times were selected to be illustrative, rather than representative.

case analysis (AUD 509,000/QALY versus AUD 233,000/QALY), in which no decrement was applied. Finally, Toumazis et al. [32] applied disutilities (ranging from 0% to 8%) to estimate the impact of indeterminate findings, yielding ICERs (for USPSTF-2013 selection criteria versus no screening) between USD 72,745 and USD 142,036/QALY.

Due to conflicting evidence, the true extent of the harms caused by screening is not known. However, as patient management and risk communication improves over time, the frequency of false positives and their psychological harms are likely to decline. Nevertheless, they can have a large impact on ICER estimates and, until accurate and context-specific data are available on the psychological harms of screening, they should be explored in sensitivity analyses. These results also highlight the benefits of investing in effective risk communication strategies for improved cost-effectiveness.

7.3. Age-adjustment

Failure to adjust for age-related changes in quality-of-life may bias cost-effectiveness analyses with lifetime time horizons. Lung cancer patients who are functionally cured, and therefore live for an extended period of time, will experience a decline in health due to ageing. The assumption that these patients retain the same quality-of-life for the remainder of their lifespan will overestimate the benefit of screening [57,63]. For example, in Fig. 1 (A and B), age-related decline is represented as a 0.05 decrement in utility occurring at ages 70 and 80. In the absence of age-adjustment (C and D), the screened individual accrues 1 additional QALY over the modelled time horizon. This bias is relatively modest but may be influential in cases where an estimated ICER falls near the willingness-to-pay threshold.

The extent to which ICERs can be biased depends on the level of age-related deterioration in quality-of-life expected. Unfortunately, age-

specific norms in the literature can be inconsistent, with some measures indicating larger declines in quality-of-life compared to others. We note that in the studies that applied age-related norms, the authors often neglected to mention the instruments selected, nor did they justify the choice of study.

In light of these issues, it is important that the utilities in an economic model are derived from the same MAUI and reflect the same valuation of health states (i.e., derived from the same scoring algorithm). Where these conditions cannot be met, the selection of substitutes should be clearly justified. Furthermore, where values from multiple studies are available, choice of inputs should be explicitly explained to minimise the risk of cherry-picking. Sensitivity analyses applying alternative values should be considered.

7.4. Lung cancer utilities

The extent to which lung cancer utilities influence ICERs depends on the other assumptions present in a model. Where models assume that early-stage lung cancer patients recover to their pre-operative condition, the key concerns should be the baseline utilities applied (e.g., Fig. 1A and 1C) and the length of the recovery period. If the time to recovery is brief, baseline utilities are more influential than lung cancer utilities. Where recovery is not assumed, lung cancer-related utilities can be influential.

The utilities applied to early-stage lung cancer patients have a larger influence on the cost-effectiveness of screening than those applied to advanced lung cancer patients, due to the short survival observed in the latter. For example, Cressman et al. [19] showed that the ICER for lung screening in Canada increased from CAD 20,724/QALY to CAD 24,392/QALY when the utility assigned to curable patients decreased by one standard error. Meanwhile, a standard error decrease in the utility for incurable patients led to an ICER of CAD 20,191/QALY. Similarly, Black et al. [14] showed that decreasing the base case utility for stage I disease by 0.04 increased the ICER for screening from USD 81,000/QALY to USD 101,000/QALY. Meanwhile, a decrease of 0.25 in the stage IV utility only reduced the ICER to USD 80,000/QALY.

Methods for calculating lung cancer utilities can also affect cost-effectiveness outcomes. Fig. 2 illustrates the consequences of using additive and multiplicative disutilities. As in Fig. 1, it compares the utility trajectories of two identical individuals diagnosed with lung cancer, one screened and the other unscreened, but they are diagnosed at an older age. To account for the lower quality-of-life at this age, a disutility for lung cancer is applied to the age-specific baseline score. Results are similar when applying the additive (Fig. 2A) and scaled multiplicative method (Fig. 2B). However, the unscaled multiplicative approach, in which disease-related utilities are multiplied with baseline utilities, causes the QALYs gained from screening to be underestimated (Fig. 2C). This bias is largely mitigated if the patient is assumed to recover.

8. Conclusion

Increasingly, health systems around the world are considering screening with LDCT as a means of improving outcomes for people with lung cancer. It is therefore important that standards for health economic evaluations are sufficient to meet the needs of policy makers. In this review, we have highlighted some key assumptions relating to quality-of-life that can impact the calculation of ICERs. All modelled evaluations of cost-effectiveness are limited by the availability of appropriate data and will always rely on assumptions to some extent. Our recommendations for future cost-effectiveness analyses are, (i) baseline utility should not be assumed to be perfect and, ideally, should reflect the smoking history of screening-eligible populations; (ii) a justifiable range of screening-related disutilities should be evaluated in sensitivity analyses unless population- and program-specific data are available; (iii) assumptions relating to post-operative recovery should be explicitly stated; (iv) the quality-of-life of lung cancer patients must be

appropriately chosen, especially if patients are not assumed to recover after curative treatment; (v) in all cases, adjustment for ageing-related changes in quality-of-life is preferable.

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Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: DJK declares honoraria from Merck Sharp & Dohme and Amgen for chairing and contributing to educational webinars. KC is co-PI of an investigator-initiated trial of cervical screening, Compass, run by the VCS Foundation, which is a government-funded not-for-profit charity; the VCS Foundation has received equipment and a funding contribution from Roche Molecular Diagnostics. She is also co-PI on a major investigator-initiated implementation program Elimination of Cervical Cancer in the Western Pacific (ECCWP) which will receive support from the Minderoo Foundation, the Frazer Family Foundation and equipment donations from Cepheid Inc. Neither KC nor her institution on her behalf receives direct funding from industry for any project. The remaining authors have no conflicts of interest to declare.

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